Final Project Based Assignment: Automated Chemical Dosing System CS3243 IoT Devices and Applications

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1 Water Level Measurement Method

Since the hospital prefers not to install any sensors inside the tank due to the periodic cleaning process, the possible way to install sensors is either on the top lid of the tank or externally at the discharge outlet point.

1.1 Configurations of the Water Supply Tank

Parameter	Value
Volume	2000 Liters
Diameter	50 cm
Height	10.16 m

Table 1: Water Supply Tank Specifications

1.2 Method : Inferential Software Flow Integration Routine

To satisfy the hospital's strict guidelines preventing any physical instrumentation from entering the 2000-liter purified water supply tank, a real-time Inferential Software Flow Integration Routine is deployed within the central controller.

This model tracks the exact volume of water entering and leaving the vessel by intercepting high-frequency square-wave pulse signals from two inline SEN-HZ21WA miniature nylon Hall Effect turbine flow sensors installed completely external to the tank housing. This product costs approximately **LKR 3000**.

Hardware Specifications

Hardware Parameter	Specification Value
Model Number	SEN-HZ21WA Miniature Water Sensor
Material Construction	Nylon PA66 Polymer with Stainless Steel Impeller Shaft
Connection Profile	1/2 inch G-Thread / DN15
Operational Flow Range	1.0 to 30.0 L/min
Excitation Voltage Input	5V DC
Sensor Mechanics	Magnetic impeller with isolated Hall Effect sensor
Signal Output Type	5V square-wave digital pulse train

Table 2: Flow Sensor Specifications

Every $\Delta t = 1.0$ second, the controller samples the background registers and calculates the discrete fluid parameters.

The current net volume of water inside the tank is updated using the following discrete Riemann summation framework.

$$V_{\text{current(new)}} = V_{\text{current(old)}} + \Delta V_{\text{net}}$$

$$\Delta V_{\text{net}} = (Q_{\text{in}} - Q_{\text{out}}) \times \Delta t$$

The instantaneous flow velocity parameters are derived in liters per minute.

$$Q_{\text{in}} = \left(\frac{\text{Inflow Pulse Count}}{\text{K-Factor}}\right) \times 60$$

$$Q_{\text{out}} = \left(\frac{\text{Outflow Pulse Count}}{\text{K-Factor}}\right) \times 60$$

The incremental volumetric deviation becomes:

$$\Delta V_{\text{net}} = \frac{\text{Inflow Pulses} - \text{Outflow Pulses}}{450}$$

This value is continuously added to the global storage variable to track active storage levels.

Converting Net Volume to Linear Height

To drive the continuous digital readout gauge on the dashboard, the controller converts the running volume variable into a linear height profile.

Given the cylindrical geometric properties of the 2000L tank:

$$r = 0.25m$$

$$A = \pi r^2$$

$$A = \pi(0.25)^2 \approx 0.19635m^2$$

The real-time height profile is calculated as:

$$V_{m^3} = \frac{V_{\text{current}}}{1000}$$

$$H = \frac{V_{m^3}}{A}$$

This dynamic variable continuously updates the HMI graphical dashboard, allowing hospital maintenance crews to clean and manually scrub the internal tank structure without risking physical damage to instrumentation.

2 Concentrated Chemical Level Estimation Method

To adhere to the strict constraint preventing the installation of level sensors inside the 10-liter concentrated chemical container, a highly reliable, non-intrusive Dual-Verification Framework is deployed.

2.1 Method: Dual-Verification (Mass-Computational) Level Tracking for Concentrated Chemical Estimation

The system is configured around a standard 10-liter cylindrical or rectangular chemical supply vessel. The target fluid handled is industrial-grade Isopropyl Alcohol (IPA), which exhibits physical properties distinct from water.

Chemical Properties and Container Specifications

- **Fluid Density (ρ_{chem}):** At room temperature ($27°C$ - $32°C$), the density of Isopropyl Alcohol is:

$$\rho_{\text{chem}} = 785 \text{ kg/m}^3$$

  which is equivalent to:

$$0.785 \text{ g/mL} = 0.785 \text{ kg/L}$$

- **Maximum Fluid Mass:**

  A completely full 10-liter chemical payload exerts a net liquid mass of:

$$M_{\text{fluid}} = \text{Volume} \times \rho_{\text{chem}}$$

$$M_{\text{fluid}} = 10 \text{ L} \times 0.785 \text{ kg/L} = 7.85 \text{ kg}$$

- **Container Tare Weight:**

  The empty plastic chemical canister exhibits a static dead weight:

$$M_{\text{tare}} \approx 0.35 \text{ kg} - 0.50 \text{ kg}$$

1. Primary Physical Verification: External Load Cell Matrix

An industrial strain-gauge load cell platform is installed directly beneath the canister assembly, isolating the electrical components entirely from chemical contact.

When a fresh container is positioned on the platform, the system filters out the weight of the platform frame and the container's dry plastic shell (M_{tare}).

The controller samples the raw live gross weight (M_{current}) and computes the true remaining physical volume (V_{mass}) via mass balance:

$$V_{\text{mass}}(\text{Liters}) = \frac{M_{\text{current}} - M_{\text{tare}}}{\rho_{\text{chem}}}$$

Because the fluid density is significantly lighter than water, this equation prevents a 21.5% overestimation error that would occur if water-density constants were mistakenly applied to the software logic.

2. Secondary Verification: Computational Flow Integration Model

Operating in parallel with the weight sensor, an inferential software loop serves as a digital twin redundancy check.

As the chemical dosing pump draws fluid from the container, it passes through an inline micro-flow meter.

Because microcontrollers execute code within sequential loops at finite clock edges, the theoretical continuous integral for fluid displacement is represented as:

$$V_{\text{dispensed}} = \int_0^t Q_{\text{chem}}(t)\, dt$$

$$\Delta t = 1.0 \text{ second}$$

The running algorithmic volume register updates dynamically every clock cycle:

$$V_{\text{algorithm(new)}} = V_{\text{algorithm(old)}} - \Delta V$$

$$\Delta V = Q_{\text{chem}} \times \Delta t$$

Where:

- Q_{chem} = Real-time volumetric flow rate extracted from the pulse frequency of the micro-flow meter (Liters/Second)
- ΔV = Absolute incremental volume drawn during the 1-second clock step (Liters)

3 Hardware Specifications and Comparative Market Analysis

This section presents a systematic evaluation of all hardware components required for the automated chemical dosing system. For each subsystem, three commercially available product options sourced from the Sri Lankan market and Foreign market are compared across cost, specifications, and suitability for the application. The recommended product for each category is highlighted in green.

3.1 Water Level Measurement - Inferential Flow Sensors

The inferential software flow integration routine requires two inline Hall Effect turbine flow sensors installed externally on the inlet and outlet pipes of the 2000-liter water supply tank. This approach eliminates any physical instrumentation inside the tank, fully satisfying the hospital's strict cleaning protocol constraint. The total cost for two sensors is approximately **LKR 6,000**.

Comparative Analysis:

Product	Supplier	Est. Cost	Key Specifications	Verdict
SEN-HZ21WA (HT30) Hall Effect sensor	duino.lk	∼LKR 3,000	1–30 L/min, 5V DC, 450 pulses/L, Nylon PA66 + stainless steel shaft, DN15	**Chosen**
G$\frac{1}{2}$ inch Hall sensor	duino.lk	∼LKR 900	1–30 L/min, 5V DC, lower-cost plastic body	**Budget alt.**
HT30 (AC-rated variant)	duino.lk	∼LKR 3,000	Same model as chosen option; identical specifications	**Duplicate**

Table 3: Water Flow Sensor Comparison (Inferential Level Method)

Justification: The SEN-HZ21WA is selected for its stainless steel impeller shaft, which provides superior corrosion resistance compared to fully plastic alternatives. At LKR 3,000 per unit, it is significantly cheaper than the commented-out ultrasonic option (LKR 65,500) and the industrial load cell option (LKR 17,479), while fully meeting the no-sensor-inside-tank constraint.

3.2 Chemical Level Estimation - Load Cell

The dual-verification framework uses a strain-gauge load cell platform mounted beneath the 10-liter IPA canister. The maximum payload is approximately 8.35 kg (7.85 kg fluid + 0.50 kg container). The load cell must accommodate this weight with adequate safety margin.

Comparative Analysis:

Product	Supplier	Est. Cost	Key Specifications	Verdict
CZL-601 60 kg S-type load cell	tronic.lk	~LKR 2,500	60 kg capacity, S-type aluminium, HX711 compatible	**Suitable**
20 kg load cell	duino.lk	~LKR 1,200	20 kg max marginal for 8.35 kg payload; minimal overload margin	**Risky**
50 kg body load cell	duino.lk	~LKR 1,800	50 kg max, ample overload headroom, HX711 compatible	**Best fit**

Table 4: Load Cell Comparison (Chemical Canister Weighing)

Justification: The 50 kg load cell provides a 6× safety margin over the maximum payload of 8.35 kg, ensuring reliable operation without risk of overload. The 20 kg option is technically borderline and unsuitable for a hospital environment where reliability is critical. The 60 kg CZL-601 functions correctly but reduces measurement resolution since the active range (0–8.35 kg) is a smaller fraction of its full scale.

3.3 Dispensing Tank Level Sensors

Each 750 mL dispensing tank requires a single low-level binary sensor to detect when the chemical level falls below the refill threshold. The sensor is fitted externally to avoid chemical contact, and only on/off detection is required — continuous level measurement is not needed for this subsystem.

Comparative Analysis:

Product	Supplier	Est. Cost	Key Specifications	Verdict
JSN-SR04T waterproof ultrasonic sensor	lk-tronics.com	~LKR 950	IP67 rated, 20–450 cm range, chemical-splash resistant, sealed probe	BestFit
HC-SR04 standard ultrasonic	tronic.lk	~LKR 350	2–400 cm range, NOT waterproof, exposed PCB — chemical exposure risk	Not Suitable
DFRobot gravity liquid sensor	dfrobot.com	~LKR 58,000	Non-contact through-wall capacitive detection, corrosion-proof, fully sealed	Budget alt.

Table 5: Dispensing Tank Level Sensor Comparison

Justification: The JSN-SR04T is the optimal choice. Its IP67 waterproof rating protects against chemical splashes in the dispensing environment, unlike the bare HC-SR04. The DFRobot capacitive sensor offers superior non-contact through-wall sensing and is the preferred option if the dispensing tanks are sealed plastic containers, at a higher cost premium.

3.4 Flow Meters - Chemical Dosing Line

The chemical dosing pump flow rate is controlled via a PWM signal but does not guarantee accurate flow. An inline flow meter on the chemical line measures the actual dispensed volume in real time, forming the secondary verification channel of the dual-verification framework.

Comparative Analysis:

Product	Supplier	Est. Cost	Key Specifications	Verdict
YF-S201 Hall effect flow sensor	lk-tronics.com	~LKR 650	1–30 L/min, 5V, Hall pulse output, ABS plastic body	**Check IPA compat.**
Magnetic plastic flow sensor	duino.lk	~LKR 600	Low flow range, food-grade plastic body	**Verify compat.**
YF-S201C G$\frac{1}{2}$ Hall effect sensor	duino.lk	~LKR 700	G$\frac{1}{2}$ thread fitting, Hall pulse output, slightly more robust pipe connection	**Good fit**

Table 6: Chemical Line Flow Meter Comparison

Important Note: IPA (Isopropyl Alcohol) can degrade standard NBR rubber O-rings and ABS seals over time. Before finalising any flow sensor for the chemical line, the impeller shaft material, O-ring compound, and housing material must be verified for IPA compatibility. PTFE or Viton-lined sensors are the recommended alternative for long-term deployments.

3.5 Control Valves - Dispensing Tank Inlets

Each dispensing tank requires a solenoid-controlled valve to regulate chemical inflow. The system may serve tens of dispensing tanks distributed throughout the hospital premises, making per-unit cost a significant factor.

Comparative Analysis:

Product	Supplier	Est. Cost	Key Specifications	Verdict
Tuya Zigbee smart gas valve	daraz.lk	~LKR 7,769	Zigbee wireless, smart-home integration, timer function, $\frac{1}{2}$–1 inch	**Overkill / costly**
FPD270A $\frac{1}{2}$ inch solenoid valve (220V)	duino.lk	~LKR 900	220V AC, $\frac{1}{2}$ inch, normally closed, direct relay control from MCU	**Best value**
4 mm pneumatic flow control valve	duino.lk	~LKR 300	Pneumatic, 4 mm tube fitting — designed for air/gas, not liquid pipelines	**Wrong type**

Table 7: Control Valve Comparison (Dispensing Tank Inlets)

Justification: With potentially tens of dispensing tanks, the FPD270A at LKR 900 per valve is the clear choice. It is driven through a relay module directly from the MCU, requires no wireless protocol stack, and is purpose-built for liquid pipelines. The Tuya Zigbee valve is designed for smart-home irrigation and gas shutoff applications and would add unnecessary cost and protocol complexity to the IoT architecture.

3.6 Variable Rate Pump - Chemical Dosing

The chemical dosing pump must deliver IPA at a controllable flow rate via PWM, with the actual flow measured by the inline flow meter. Since IPA is a chemical solvent, fluid compatibility with the pump internals is the primary selection criterion.

Comparative Analysis:

Product	Supplier	Est. Cost	Key Specifications	Verdict
100 W submersible water pump	duino.lk	~LKR 3,500	PWM-controllable, 4.5 m head — designed for water, chemical compatibility unverified	△ Compat. risk
Peristaltic liquid dosing pump	duino.lk	~LKR 2,800	Fluid isolated inside silicone/rubber tube; pump body never contacts IPA; PWM speed control	Best for chemicals
220V 1000 L/h submersible pump	duino.lk	~LKR 1,800	1000 L/h flow rate — far exceeds dosing requirement for 750 mL dispensing tanks	Flow rate too high

Table 8: Chemical Dosing Pump Comparison

Justification: The peristaltic dosing pump is the architecturally correct choice for this application. In a peristaltic design, the pumped fluid travels entirely inside a replaceable tube, meaning the IPA never contacts the pump motor, housing, or impeller assembly. This eliminates chemical compatibility concerns entirely. PWM speed control is also more linear for peristaltic pumps than centrifugal designs, improving dosing accuracy.

3.7 Summary Cost Comparison

Table 9 consolidates the recommended products and their estimated unit costs.

Subsystem	Selected Product	Unit Cost (LKR)	Qty (min.)
Water level — flow sensor	SEN-HZ21WA (HT30)	∼3,000	2
Chemical level — load cell	50 kg body load cell	∼1,800	1
Dispensing tank sensor	JSN-SR04T ultrasonic	∼950	n tanks
Chemical line flow meter	YF-S201C G$\frac{1}{2}$	∼700	1
Control valve (per tank)	FPD270A solenoid	∼900	n tanks
Dosing pump	Peristaltic pump	∼2,800	1

Table 9: Recommended Component Cost Summary

The minimum fixed hardware cost (excluding dispensing-tank-count-dependent components) is approximately **LKR 14,300**. Each additional dispensing tank requires one JSN-SR04T sensor (∼LKR 950) and one FPD270A solenoid valve (∼LKR 900), totalling **LKR 1,850 per tank**.